1. Description

1.1. Project

Project Name	MyApplication
Board Name	custom
Generated with:	STM32CubeMX 6.17.0
Date	04/11/2026

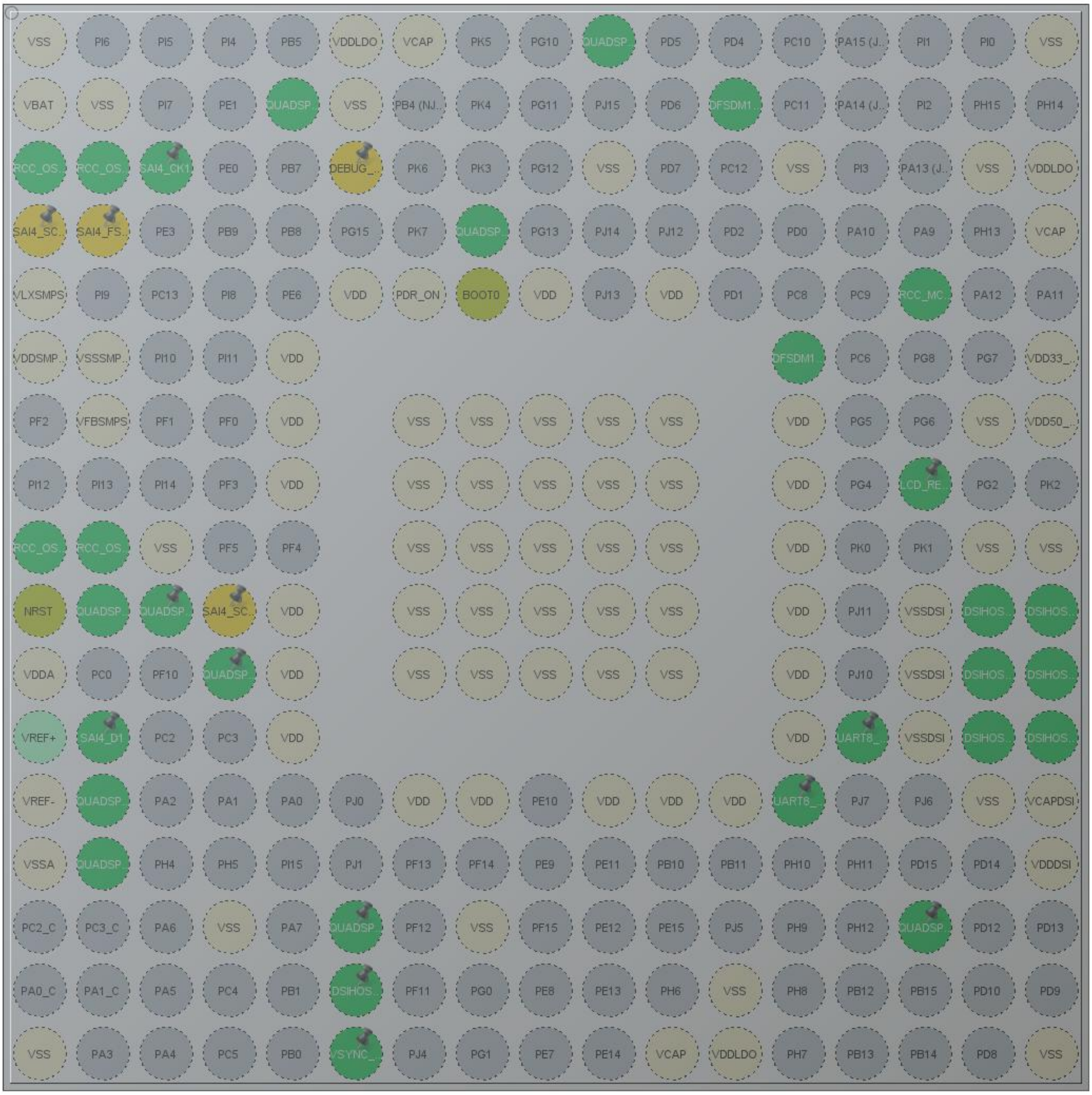
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H747/757
MCU name	STM32H747XIHx
MCU Package	TFBGA240
MCU Pin number	265

1.3. Core(s) information

Core(s)	ARM Cortex-M7 ARM Cortex-M4

2. Pinout Configuration



TFBGA240 +25 (Top view)

3. Pins Configuration

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSS	Power		
A6	VDDLDO	Power		
A7	VCAP	Power		
A10	PG9	I/O	QUADSPI_BK2_IO2	
A17	VSS	Power		
B1	VBAT	Power		
B2	VSS	Power		
B5	PB6	I/O	QUADSPI_BK1_NCS	
B6	VSS	Power		
B12	PD3	I/O	DFSDM1_CKOUT	
C1	PC15-OSC32_OUT (OSC32_OUT)	I/O	RCC_OSC32_OUT	
C2	PC14-OSC32_IN (OSC32_IN)	I/O	RCC_OSC32_IN	
C3	PE2	I/O	SAI4_CK1	
C6	PB3 (JTDO/TRACESWO) *	I/O	DEBUG_JTDO-SWO	
C10	VSS	Power		
C13	VSS	Power		
C16	VSS	Power		
C17	VDDLDO	Power		
D1	PE5 *	I/O	SAI4_SCK_A	
D2	PE4 *	I/O	SAI4_FS_A	
D8	PG14	I/O	QUADSPI_BK2_IO3	
D17	VCAP	Power		
E1	VLXSMPS	Power		
E6	VDD	Power		
E7	PDR_ON	Power		
E8	BOOT0	Boot		
E9	VDD	Power		
E11	VDD	Power		
E15	PA8	I/O	RCC_MCO_1	
F1	VDDSMPS	Power		
F2	VSSSMPS	Power		
F5	VDD	Power		
F13	PC7	I/O	DFSDM1_DATIN3	
F17	VDD33_USB	Power		

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
G2	VFBSMPS	Power		
G5	VDD	Power		
G7	VSS	Power		
G8	VSS	Power		
G9	VSS	Power		
G10	VSS	Power		
G11	VSS	Power		
G13	VDD	Power		
G16	VSS	Power		
G17	VDD50_USB	Power		
H5	VDD	Power		
H7	VSS	Power		
H8	VSS	Power		
H9	VSS	Power		
H10	VSS	Power		
H11	VSS	Power		
H13	VDD	Power		
H15	PG3 **	I/O	GPIO_Output	LCD_RESET
J1	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
J2	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
J3	VSS	Power		
J7	VSS	Power		
J8	VSS	Power		
J9	VSS	Power		
J10	VSS	Power		
J11	VSS	Power		
J13	VDD	Power		
J16	VSS	Power		
J17	VSS	Power		
K1	NRST	Reset		
K2	PF6	I/O	QUADSPI_BK1_IO3	
K3	PF7	I/O	QUADSPI_BK1_IO2	
K4	PF8 *	I/O	SAI4_SCK_B	
K5	VDD	Power		
K7	VSS	Power		
K8	VSS	Power		
K9	VSS	Power		
K10	VSS	Power		
K11	VSS	Power		

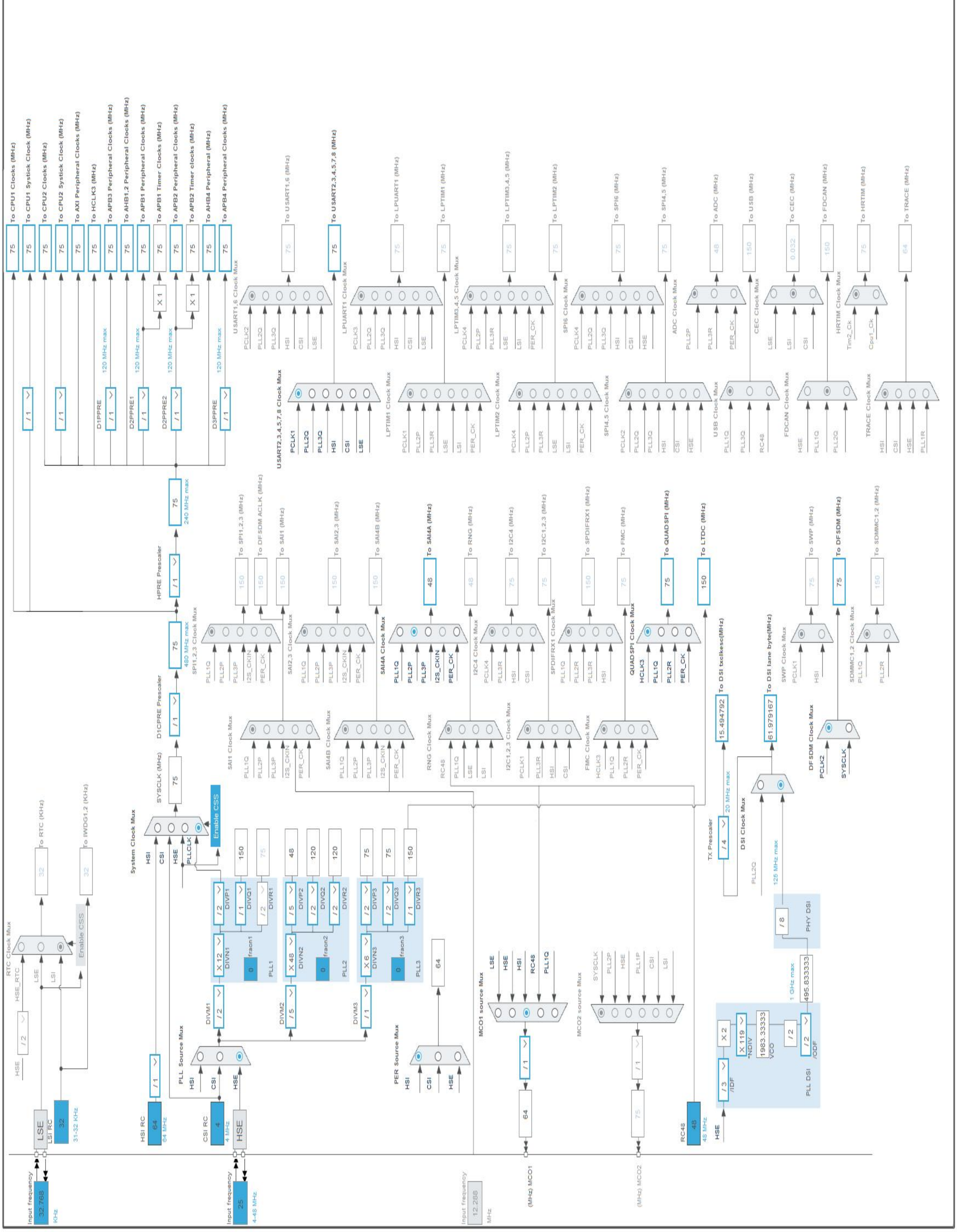
Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
K13	VDD	Power		
K15	VSSDSI	Power		
K16	DSI_D1P	MonoIO	DSIHOST_D1P	
K17	DSI_D1N	MonoIO	DSIHOST_D1N	
L1	VDDA	Power		
L4	PF9	I/O	QUADSPI_BK1_IO1	
L5	VDD	Power		
L7	VSS	Power		
L8	VSS	Power		
L9	VSS	Power		
L10	VSS	Power		
L11	VSS	Power		
L13	VDD	Power		
L15	VSSDSI	Power		
L16	DSI_CKP	MonoIO	DSIHOST_CKP	
L17	DSI_CKN	MonoIO	DSIHOST_CKN	
M2	PC1	I/O	SAI4_D1	
M5	VDD	Power		
M13	VDD	Power		
M14	PJ9	I/O	UART8_RX	
M15	VSSDSI	Power		
M16	DSI_D0P	MonoIO	DSIHOST_D0P	
M17	DSI_D0N	MonoIO	DSIHOST_D0N	
N1	VREF-	Power		
N2	PH2	I/O	QUADSPI_BK2_IO0	
N7	VDD	Power		
N8	VDD	Power		
N10	VDD	Power		
N11	VDD	Power		
N12	VDD	Power		
N13	PJ8	I/O	UART8_TX	
N16	VSS	Power		
N17	VCAPDSI	Power		
P1	VSSA	Power		
P2	PH3	I/O	QUADSPI_BK2_IO1	
P17	VDDDSI	Power		
R4	VSS	Power		
R6	PB2	I/O	QUADSPI_CLK	
R8	VSS	Power		

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
R15	PD11	I/O	QUADSPI_BK1_IO0	
T6	PJ2	I/O	DSIHOST_TE	
T12	VSS	Power		
U1	VSS	Power		
U6	PJ3 **	I/O	GPIO_Output	VSYNC_FREQ
U11	VCAP	Power		
U12	VDDLDO	Power		
U17	VSS	Power		

** The pin is affected with an I/O function
 * The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration

RIF configuration not available for STM32H747XIHx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H747/757
MCU	STM32H747XIHx
Datasheet	DS12930_Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(DD36000)
Capacity	36000.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	450.0 mA
Max Pulse Current	1000.0 mA
Cells in series	1
Cells in parallel	1

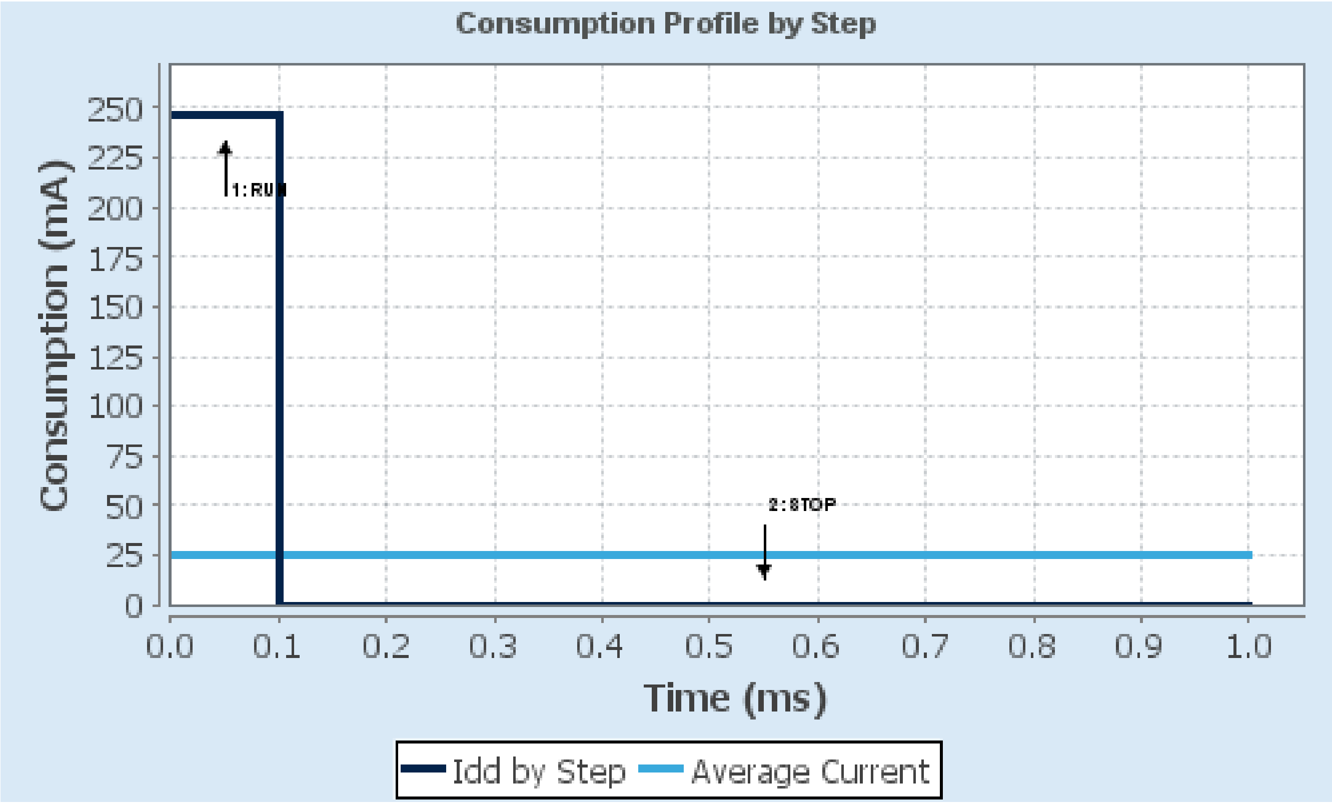
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN/CRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	CM7: ITCM/Cache / CM4: FLASH_B/ART	CM7: NA / CM4: NA
CM7 Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL ALL_IPs_ON	LSE Flash-ON
CM4 Frequency	240 MHz	0 Hz
Clock Source Frequency	25 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	247 mA	145 µA
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	24.83 mA
Battery Life	1 month, 29 days, 21 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
250
225
200
175
150
125
100
75
50
25
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	MyApplication
Project Folder	C:\TouchGFXProjects\MyApplication
Toolchain / IDE	EWARM V8.50
Firmware Package Name and Version	STM32Cube FW_H7 V1.13.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	M4-0x1000
Minimum Stack Size	M4-0x1000

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls ARM Cortex-M7

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_MDMA_Init	MDMA
4	MX_QUADSPI_Init	QUADSPI
5	MX_DMA2D_Init	DMA2D
6	MX_DSIHOST_DSI_Init	DSIHOST
7	MX_LTDC_Init	LTDC
8	MX_FREERTOS_Init	FREERTOS_M7
9	MX_CRC_Init	CRC
10	MX_JPEG_Init	JPEG
11	MX_LIBJPEG_Init	LIBJPEG

Rank	Function Name	Peripheral Instance Name
12	MX_SAI4_Init	SAI4
13	MX_UART8_Init	UART8
14	MX_DFSDM1_Init	DFSDM1
16	MX_TouchGFX_Init	STMicroelectronics.X-CUBE-TOUCHGFX.4.26.1_M7
17	MX_TouchGFX_Process	STMicroelectronics.X-CUBE-TOUCHGFX.4.26.1_M7

2.4. Advanced Settings - Generated Function Calls ARM Cortex-M4

Rank	Function Name	Peripheral Instance Name
1	MX_MDMA_Init	MDMA

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7

3.1.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Speculation default mode Settings:

Speculation default mode	Disabled

Cortex Interface Settings:

CPU ICache	**Enabled ***
CPU DCache	**Enabled ***

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	**Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers ***

Cortex Memory Protection Unit Region 0 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x90000000 ***
MPU Region Size	**256MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x90000000 ***
MPU Region Size	**128MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE

MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0xD0000000 ***
MPU Region Size	**32MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x24000000 ***
MPU Region Size	**512KB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	ENABLE
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x10000000 ***
MPU Region Size	**256KB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	ENABLE
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x10040000 ***

MPU Region Size	**32KB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region	Disabled

3.2. CRC
mode: Activated

3.2.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	

	D3

Basic Parameters:

Default Polynomial State	Enable
Default Init Value State	Enable

Advanced Parameters:

Input Data Inversion Mode	None
Output Data Inversion Mode	Disable
Input Data Format	Bytes

3.3. DFSDM1
mode: PDM/SPI input from ch3 and internal clock
3.3.1. Filter 0:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

regular channel selection:

regular channel selection	- None -

injected channel selection:

Channel0 as injected channel	Disable
Channel1 as injected channel	Disable
Channel2 as injected channel	Disable
Channel3 as injected channel	Disable
Channel4 as injected channel	Disable
Channel5 as injected channel	Disable
Channel6 as injected channel	Disable
Channel7 as injected channel	Disable

3.3.2. Filter 1:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

regular channel selection:

regular channel selection	- None -

injected channel selection:

Channel0 as injected channel	Disable
Channel1 as injected channel	Disable
Channel2 as injected channel	Disable
Channel3 as injected channel	Disable
Channel4 as injected channel	Disable
Channel5 as injected channel	Disable
Channel6 as injected channel	Disable
Channel7 as injected channel	Disable

3.3.3. Filter 2:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

regular channel selection:

regular channel selection	- None -

injected channel selection:

Channel0 as injected channel	Disable
Channel1 as injected channel	Disable
Channel2 as injected channel	Disable
Channel3 as injected channel	Disable
Channel4 as injected channel	Disable
Channel5 as injected channel	Disable
Channel6 as injected channel	Disable
Channel7 as injected channel	Disable

3.3.4. Filter 3:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

regular channel selection:

regular channel selection	- None -

injected channel selection:

Channel0 as injected channel	Disable
Channel1 as injected channel	Disable
Channel2 as injected channel	Disable
Channel3 as injected channel	Disable
Channel4 as injected channel	Disable
Channel5 as injected channel	Disable
Channel6 as injected channel	Disable
Channel7 as injected channel	Disable

3.3.5. Output Clock:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Output Clock parameters:

Selection	Source for ouput clock is system clock
Divider	2

3.3.6. Channel 3:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Channel 3 parameters:

Type	SPI with rising edge
Spi Clock	Internal SPI clock
Offset	0
Right Bit Shift	**0x00 ***

Analog watchdog parameters:

Filter Order	FastSinc filter type
Oversampling	1

3.4. DMA2D

mode: Activated

3.4.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Basic Parameters:

Transfer Mode	**Register to Memory ***
Color Mode	**RGB888 ***
Output Offset	0

3.5. DSIHOST

DSIHost: Adapted Command Mode with TE Pin

3.5.1. DSI Clocks:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

from PLLDSI:

Pllndiv	119
Pllodf	DSI_PLL_OUT_DIV2
Pllidf	DSI_PLL_IN_DIV3
High Speed Clock - PLLDSI Output	495833
Lane Byte Clock	61979
Tx Escape Ckdiv	4
Transmission Escape Clock	15495
Time Out Clock	61979

from PLL2Q:

Lane Byte Clock	120000
Transmission Escape Clock	30000
Time Out Clock	120000

3.5.2. Timeout Counters:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Time Out Clock Setting:

Time Out Clock Divider	1
Time Out Clock - from PLLDSI	61979

Contention Error Detection:

High-speed transmission time-out	0
No High-speed transmission time-out	
Low-power reception time-out	0
No Low-power reception time-out	

Peripheral Response:

High-speed read time-out	0
Resulting Timing High-speed read time-out	0
Low-power read time-out	0
Resulting Timing Low-power read time-out	0
High-speed write time-out	0
Resulting Timing High-speed write time-out	0
Low-power write time-out	0
Resulting Timing Low-power write time-out	0
BTA time-out	0
Resulting Timing for BTA time-out	0
High-speed write presp mode	Normal Behavior

3.5.3. Data and Clock Lanes:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Basic Settings:

Number of Lanes	**Two Data Lanes ***
Automatic Clock Lane Control	Clock lane is always provided
Bus Turn Around Request is	Enabled

Flow Control - Configuration:

CRC Reception	Disabled

ECC Reception	Disabled
EoTP Reception is	Disabled
EoTP Transmission is	Disabled
Acknowledge Request after Each Transmission	**Enabled ***
Tearing Effect Acknowledge Request Enable	**Enabled ***

Flow Control - Packet Analyzer Configuration:

CRC Error Interrupt	Disable
ECC Errors Interrupt	Disable
EoTP Error Interrupt	Disable
Packet Size Error Interrupt	Disable
Acknowledge Errors Interrupt	Disable
PHY related Errors Interrupt	Disable

3.5.4. PHY Timings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

LP to HS and HS to LP Transitions Timings:

Minimum wait period to request a HS transmission after the Stop state (min est 1 cycle de escape clock)	0
Resulting Minimum wait period to request a HS transmission after the Stop state (min est 1 cycle de escape clock)	0

3.5.5. Commands:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

APB Interface Error Configuration:

Generic Command Error Interrupt	Disable

Transmission Mode for Commands::

Generic Short Write Zero Parameter	**Low Power Transmission ***
Generic Short Write One Parameter	**Low Power Transmission ***
Generic Short Write Two parameters	**Low Power Transmission ***

Generic Short Read Zero parameter	**Low Power Transmission ***
Generic Short Read One parameter	**Low Power Transmission ***
Generic Short Read Two parameters	**Low Power Transmission ***
Generic Long Write	**Low Power Transmission ***
DCS Short Write Zero parameter	**Low Power Transmission ***
DCS Short Write One parameter	**Low Power Transmission ***
DCS Short Read Zero parameter	**Low Power Transmission ***
DCS Long Write	**Low Power Transmission ***
Maximum Read Packet Size Command	**Low Power Transmission ***

3.5.6. Display Interface:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Basic Settings:

Display ID	0
Color Coding	RGB888 (24 bits) - DSI mode

Specific Command Mode Settings:

Maximum Command Size	400
The Refresh of the Display Frame Buffer is Triggered	**manually by Enabling the LTDC ***
Tearing Effect Source	External Source
Polarity of the External Tearing Effect Source	Rising Edge

3.5.7. LTDC Interface:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Clocking:

LTDC Pixel Clock (from Clock tree)	150

Frame Vertical Timings:

VSA: Vertical Synchronism Active duration (set in LTDC)	1

VBP: Vertical Back-Porch duration (set in LTDC)	2
VFP: Vertical Front-Porch duration (set in LTDC)	1
VACT: Vertical Active duration (set in LTDC)	480
Frame Horizontal Timings:	
HSA: Horizontal Synchronism Active duration (set in LTDC)	1
HBP: Horizontal Back-Porch duration (set in LTDC)	2
HACT: Horizontal Active duration (set in LTDC)	400
HLINE: Horizontal Line duration (set in LTDC)	404
Polarity of the Control Signals:	
VSYNC Polarity (set in LTDC)	DSI_VSYNC_ACTIVE_HIGH
HSYNC Polarity (set in LTDC)	DSI_HSYNC_ACTIVE_HIGH
Data Enable Polarity (set in LTDC)	DSI_DATA_ENABLE_ACTIVE_HIGH
Interface:	
The DSI is halting the LTDC synchronously	On a Rising Edge of VSYNC
Interface Error Configuration:	
LTDC FIFO Overflow Error Interrupt is	Disabled

3.6. JPEG
mode: Activated
3.6.1. Parameter Settings:

Core(s) Settings:	
Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1
Version:	
JPEG version	jpeg1_v1_0
JPEG Software options:	
ENCODE	Enabled
DECODE	Enabled
RGB_FORMAT	**JPEG_RGB888 ***
JPEG_SWAP	RG

3.7. LTDC

Display Type: RGB888 (24 bits) - DSI mode

3.7.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Synchronization for Width:

Horizontal Synchronization Width	**1 ***
Horizontal Back Porch	**2 ***
Active Width	**400 ***
Horizontal Front Porch	**1 ***
HSync Width	0
Accumulated Horizontal Back Porch Width	2
Accumulated Active Width	402
Total Width	403

Synchronization for Height:

Vertical Synchronization Height	**1 ***
Vertical Back Porch	2
Active Height	480
Vertical Front Porch	**1 ***
VSync Height	0
Accumulated Vertical Back Porch Height	2
Accumulated Active Height	482
Total Height	483

Signal Polarity:

Horizontal Synchronization Polarity	**Active High ***
Vertical Synchronization Polarity	**Active High ***
Data Enable Polarity	Active Low
Pixel Clock Polarity	Normal Input

Layer Default Color:

Red	0
Green	0
Blue	0

3.7.2. Layer Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Layer Default Color:

Layer 0 - Alpha	0
Layer 0 - Blue	0
Layer 0 - Green	0
Layer 0 - Red	0

Number of Layers:

Number of Layers	**1 layer ***

Windows Position:

Layer 0 - Window Horizontal Start	0
Layer 0 - Window Horizontal Stop	**400 ***
Layer 0 - Window Vertical Start	0
Layer 0 - Window Vertical Stop	**480 ***

Pixel Parameters:

Layer 0 - Pixel Format	**RGB888 ***

Blending:

Layer 0 - Alpha constant for blending	**255 ***
Layer 0 - Blending Factor1	**Alpha constant x Pixel Alpha ***
Layer 0 - Blending Factor2	**Alpha constant x Pixel Alpha ***

Frame Buffer:

Layer 0 - Color Frame Buffer Start Adress	**0xD0000000 ***
Layer 0 - Color Frame Buffer Line Length (Image Width)	**400 ***
Layer 0 - Color Frame Buffer Number of Lines (Image Height)	**480 ***

3.8. QUADSPI

QuadSPI Mode: Dual Bank with Quad Lines

Chip Select for Dual bank: Enable Chip Select 1 for both banks

3.8.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7

Power Domain:	D1

General Parameters:

Clock Prescaler	**3 ***
Fifo Threshold	1
Sample Shifting	No Sample Shifting
Flash Size	1
Chip Select High Time	1 Cycle
Clock Mode	Low
Dual Flash	Enabled

3.9. RCC
High Speed Clock (HSE): Crystal/Ceramic Resonator
Low Speed Clock (LSE) : Crystal/Ceramic Resonator
mode: Master Clock Output 1
3.9.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	D3

Power Parameters:

SupplySource	PWR_DIRECT_SMPS_SUPPLY
Power Regulator Voltage Scale	**Power Regulator Voltage Scale 1 ***

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	1 WS (2 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 8 and 16 MHz
PLL2 input frequency range	Between 4 and 8 MHz

PLL3 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	MEDIUM VCO range
PLL2 clock Output range	Wide VCO range
PLL3 clock Output range	MEDIUM VCO range

3.10. SAI4
Mode: Pulse Density Modulation using MicroPhones pairs (1-2) and CK1 as clock
3.10.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D3

SAI A:

Synchronization Inputs	Asynchronous
Mic Pairs Nbr	1
Clock Enable	Pdm Clock1 Enable
Protocol	Free
Audio Mode	Master Receive
Frame Length (only Even Values)	8
Data Size	8 Bits
Slot Size	DataSize
Output Mode	Stereo
Companding Mode	No companding mode
First Bit	MSB First
Frame Synchro Active Level Length	1
Frame Synchro Definition	Start Frame
Frame Synchro Polarity	Active Low
Frame Synchro Offset	First Bit
First Bit Offset	0
Number of Slots	1
Slot Active Final Value	0x00000000
Slot Active	Neither
Clock Source	SAI PLL Clock
Master Clock No Divider	Enabled
Audio Frequency	192 KHz
Real Audio Frequency	**187.5 KHz ***
Error between Selected	**-2.34 % ***
Clock Strobing	Falling Edge

Fifo Threshold	Empty
Output Drive	Disabled
Master Clock Over Sampling	Disabled

3.11. SYS_M4
Timebase Source: SysTick
3.11.1. Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	

3.12. SYS
Timebase Source: TIM6
3.12.1. Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	

3.13. UART8
Mode: Asynchronous
3.13.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Basic Parameters:

Baud Rate	**921600 ***
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.14. FREERTOS_M7
Interface: CMSIS_V2
3.14.1. Config parameters:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

API:

FreeRTOS API	CMSIS v2

Versions:

FreeRTOS version	10.6.2
CMSIS-RTOS version	2.1.3

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000

MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	**Enabled ***
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	**100000 ***
HEAP_CLEAR_MEMORY_ON_FREE	0
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	**Enabled ***
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.14.2. Include parameters:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.14.3. Advanced settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled

Project settings (see parameter description first):

Use FW pack heap file	Enabled

3.15. LIBJPEG
mode: Enabled

3.15.1. Config parameters:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Version:

LIBJPEG version	8d

MW configuration:

Data Stream management type	Stdio
FREERTOS	Enabled
HAVE_BOOLEAN	Undefined

General Settings:

Use FREERTOS Memory Allocator	Enabled

3.16. STMicroelectronics.X-CUBE-TOUCHGFX.4.26.1_M7
mode: GraphicsJjApplication

3.16.1. TouchGFX Generator:

Core(s) Settings:

Context(s):	Cortex-M7

Initialized Context:	Cortex-M7
Power Domain:	D1

Display:

Interface	Custom
Framebuffer Pixel Format	**RGB888 ***
Width	**800 ***
Height	**480 ***
Use Larger Framebuffer Stride	No
Framebuffer Strategy	Single Buffer
Buffer Location	By Allocation

Driver:

Application Tick Source	Custom
Use DMA2D Accelerator (ChromART)	**Yes ***
Real-Time Operating System	CMSIS_RTOS_V2

Additional Features:

Vector Rendering	**Software ***
Vector Font Rendering	**Enabled ***

Video Decoding:

Type	**Hardware ***
Concurrent videos	1
Strategy	**Single buffer ***
Buffer Width	**800 ***
Buffer Height	**480 ***
Orientation	Native

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
DFSDM1	PD3	DFSDM1_CKOUT	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PC7	DFSDM1_DATIN3	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
DSIHOST	DSI_D1P	DSIHOST_D1P	n/a	n/a	n/a		Cortex-M7	D1
	DSI_D1N	DSIHOST_D1N	n/a	n/a	n/a		Cortex-M7	D1
	DSI_CKP	DSIHOST_CKP	n/a	n/a	n/a		Cortex-M7	D1
	DSI_CKN	DSIHOST_CKN	n/a	n/a	n/a		Cortex-M7	D1
	DSI_D0P	DSIHOST_D0P	n/a	n/a	n/a		Cortex-M7	D1
	DSI_D0N	DSIHOST_D0N	n/a	n/a	n/a		Cortex-M7	D1
	PJ2	DSIHOST_TE	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
QUADSPI	PG9	QUADSPI_BK2_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PB6	QUADSPI_BK1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PG14	QUADSPI_BK2_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PF6	QUADSPI_BK1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PF7	QUADSPI_BK1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PF9	QUADSPI_BK1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PH2	QUADSPI_BK2_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PH3	QUADSPI_BK2_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PB2	QUADSPI_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
	PD11	QUADSPI_BK1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D1
RCC	PC15-OSC32_	RCC_OSC32_OUT	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
	OUT							
	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
	PA8	RCC_MCO_1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	D3
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
SAI4	PE2	SAI4_CK1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***		Cortex-M7	D3
	PC1	SAI4_D1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D3
UART8	PJ9	UART8_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PJ8	UART8_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
Single Mapped Signals	PB3 (JTDO/TRACESWO	DEBUG_JTDO-SWO	n/a	n/a	n/a			
	PE5	SAI4_SCK_A	Alternate Function Push Pull	No pull-up and no pull-down	Low			
	PE4	SAI4_FS_A	Alternate Function Push Pull	No pull-up and no pull-down	Low			
	PF8	SAI4_SCK_B	Alternate Function Push Pull	No pull-up and no pull-down	Low			
GPIO	PG3	GPIO_Output	Output Push Pull	**Pull-up ***	**Very High ***	LCD_RESET	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	VSYNC_FREQ	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

* Initialized context

4.2. DMA configuration

nothing configured in DMA service

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

MDMA Request	Channel	Trigger Mode	Priotity
JPEG_INFIFO_TH	MDMA_Channel7	Buffer transfer	**High ***
JPEG_OUTFIFO_TH	MDMA_Channel6	Buffer transfer	**Very High ***

JPEG_INFIFO_TH: MDMA_Channel7 DMA request Settings:

Buffer Transfer Length:	**32 ***
Data Mask:	0
Source Burst Size:	**Burst of 32 beats ***
Destination Burst Size:	**Burst of 16 beats ***
Block Data Length:	0
Destination Address:	0
Destination Block Address Offset:	0
Address Mask:	0
Data Alignement:	N/A
Source Block Address Offset:	0
Endianness:	Little endianness preserved
Source Address:	0
Source Increment mode:	Increment by a byte
Block Count:	0
Destination Increment mode:	Fixed
Source Data Width:	Byte
Destination Data Width:	**Word ***

JPEG_OUTFIFO_TH: MDMA_Channel6 DMA request Settings:

Buffer Transfer Length:	**32 ***
Data Mask:	0
Source Burst Size:	**Burst of 32 beats ***
Destination Burst Size:	**Burst of 32 beats ***
Block Data Length:	0
Destination Address:	0
Destination Block Address Offset:	0

Address Mask:	0
Data Alignement:	N/A
Source Block Address Offset:	0
Endianness:	Little endianness preserved
Source Address:	0
Source Increment mode:	Fixed
Block Count:	0
Destination Increment mode:	Increment by a byte
Source Data Width:	**Word ***
Destination Data Width:	Byte

4.5. NVIC configuration

4.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	true	0	0
LTDC global interrupt	true	7	0
DMA2D global interrupt	true	7	0
JPEG global interrupt	true	5	0
MDMA global interrupt	true	5	0
DSI global Interrupt	true	7	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
CM4 send event interrupt for CM7	unused		
FPU global interrupt	unused		
UART8 global interrupt	unused		
LTDC global error interrupt	unused		
QUADSPI global interrupt	unused		
DFSDM1 filter0 global interrupt	unused		
DFSDM1 filter1 global interrupt	unused		
DFSDM1 filter2 global interrupt	unused		
DFSDM1 filter3 global interrupt	unused		
HSEM1 global interrupt	unused		
SAI4 global interrupt	unused		
Hold core interrupt	unused		

4.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	false	true	true
LTDC global interrupt	false	true	true
DMA2D global interrupt	false	true	true
JPEG global interrupt	false	true	true
MDMA global interrupt	false	true	true
DSI global Interrupt	false	true	true

4.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
CM7 send event interrupt for CM4	unused		
FPU global interrupt	unused		
HSEM2 global interrupt	unused		
Hold core interrupt	unused		

4.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false

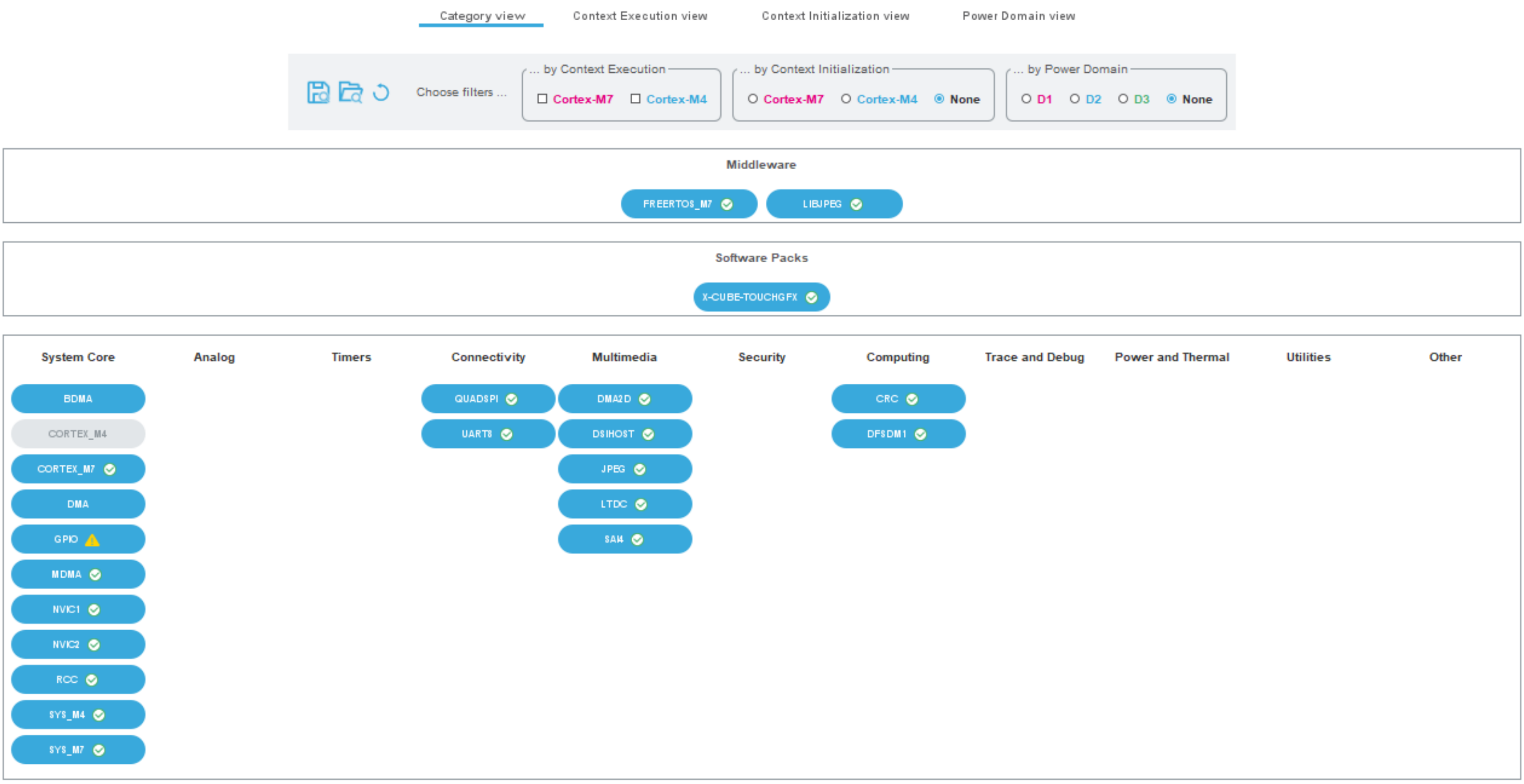
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

*** User modified value**

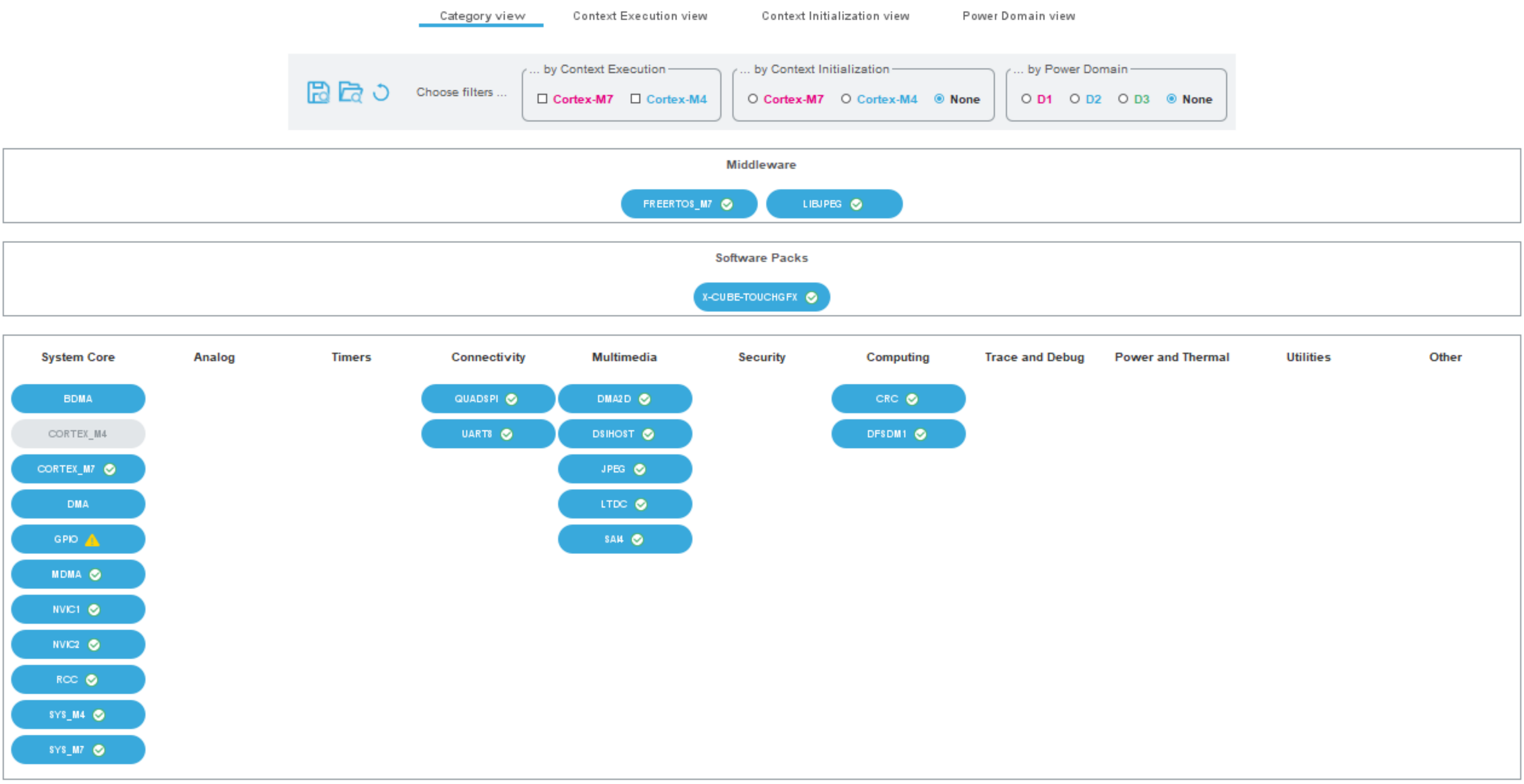
5. System Views

5.1. Category view

5.1.1. Current



5.1.2. Without filters



5.2. Context Execution view

5.3. Context Initialization view

5.4. Power Domain view

6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
STMicroelectronics	X-CUBE-TOUCHGFX	4.26.1	Class : Graphics Group : Application Variant : TouchGFX Generator Version : 4.26.1

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-

	guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5286-stm32h7x5x7-dualcore-microcontroller-debugging-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5557-stm32h745755-and-stm32h747757-lines-dualcore-architecture-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5617-stm32h745755-and-stm32h747757-lines-interprocessor-communications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4860-introduction-to-dsi-host-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
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